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Machine Learning for Shock-Capturing Numerical Schemes

Learning reconstruction decisions for
1D hyperbolic conservation laws

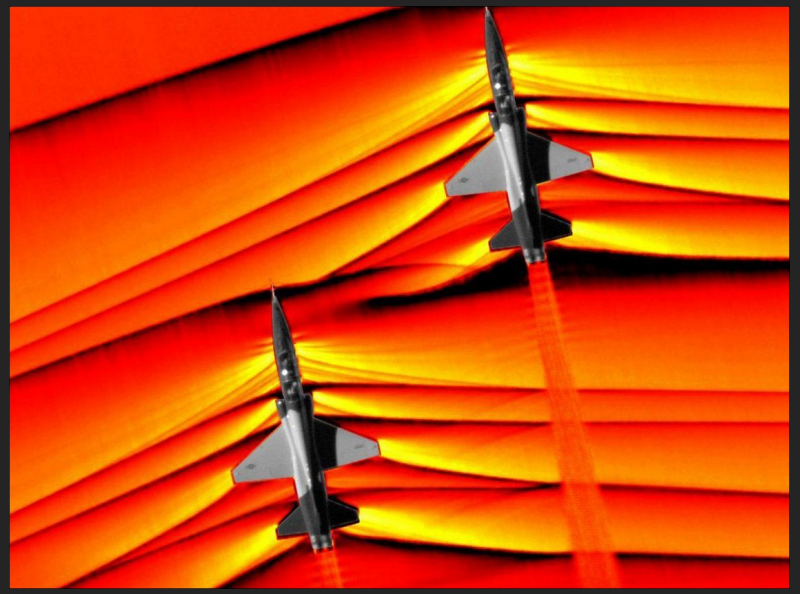
WENO-NN / NN-TENO5 / DIFFERENTIABLE CFD

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PHYSICS × NUMERICS × LEARNING



Capturing sharp fronts without spurious oscillations

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0$$